

The empirical recurrence for OEIS A252062, recovered from its tail

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Abstract

OEIS A252062 records “Empirical recurrence of order 98” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 180$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 104$. Its last coefficient is $c_{98} = -80 \neq 0$, so no further tail satisfies anything shorter; any order-98 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A252062 is “Number of $(n+2) \times (5+2)$ 0..2 arrays with every 3×3 subblock row, column, diagonal and antidiagonal sum not equal to 1 2 or 4.”. It has offset 1 and begins

1948, 5610, 25220, 126936, 664938, 3510697, 18508976, 97654871, ...

The entry states:

Empirical recurrence of order 98 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 10:44:44, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 180$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 180$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 98: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 98.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 360$ exact terms from $n = 6$ on, it returns order 98 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{98} c_i a(n-i),$$

c_1	7	c_{26}	−1346579501	c_{51}	86634904335	c_{76}	67691042
c_2	−17	c_{27}	−1607711485	c_{52}	88333307093	c_{77}	−37734770
c_3	75	c_{28}	205094196	c_{53}	43969802753	c_{78}	−30410735
c_4	−126	c_{29}	3792788297	c_{54}	−17228361373	c_{79}	15959841
c_5	−123	c_{30}	6276206077	c_{55}	−43562900203	c_{80}	28393826
c_6	−734	c_{31}	3136983516	c_{56}	−34142706865	c_{81}	11032083
c_7	−1984	c_{32}	−6903719395	c_{57}	−5909200952	c_{82}	−8544453
c_8	7759	c_{33}	−17140852220	c_{58}	14599543323	c_{83}	−13223247
c_9	16837	c_{34}	−16300753786	c_{59}	14864450134	c_{84}	−6571895
c_{10}	32802	c_{35}	3979001775	c_{60}	4144180135	c_{85}	142303
c_{11}	−23326	c_{36}	32051447301	c_{61}	−7441189109	c_{86}	2622425
c_{12}	−292463	c_{37}	46425431640	c_{62}	−9574949696	c_{87}	1618353
c_{13}	−486025	c_{38}	21064077489	c_{63}	−5427929137	c_{88}	332815
c_{14}	−298569	c_{39}	−31945132075	c_{64}	1245027403	c_{89}	−251823
c_{15}	1921402	c_{40}	−82383852131	c_{65}	4850560403	c_{90}	−207349
c_{16}	5775830	c_{41}	−77156675386	c_{66}	5196228832	c_{91}	−74153
c_{17}	5526168	c_{42}	−14271871351	c_{67}	2996671606	c_{92}	−10469
c_{18}	−6340168	c_{43}	78584466593	c_{68}	480162476	c_{93}	−877
c_{19}	−36477866	c_{44}	125628639713	c_{69}	−1354441105	c_{94}	−2370
c_{20}	−56926077	c_{45}	96249601023	c_{70}	−1926233123	c_{95}	−894
c_{21}	−7729429	c_{46}	−5074204009	c_{71}	−1458960456	c_{96}	−474
c_{22}	151654666	c_{47}	−99571200039	c_{72}	−569277200	c_{97}	−84
c_{23}	336323472	c_{48}	−135605531829	c_{73}	179987633	c_{98}	−80
c_{24}	252198876	c_{49}	−78817454348	c_{74}	415008611		
c_{25}	−392242436	c_{50}	10836752629	c_{75}	297964223		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 104$, and it is the recurrence OEIS A252062 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{98} - c_1x^{98-1} - \dots - c_{98}$. Its constant term is $-c_{98} = 80$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 98 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 98, so $\deg q = 98$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 98, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 98, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -80 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-98 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A252062.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.